

Particles suspended in a fluid move in an apparently random fashion in what is known as Brownian motion (named after the botanist Robert Brown, who observed the motion of pollen grains in water in 1827). This can be understood by treating the fluid as composed of many molecules and then treating the motion of these molecules in the fluid statistically. In most of our discussions we have related statistical mechanics to thermodynamics, by using microscopic degrees of freedom to calculate some statistical quantity, e.g. Ω in the microcanonical ensemble, or Z in the canonical ensemble, and then connected these quantities to thermodynamic variables, such as entropy or the Helmholtz free energy. Once we have made a connection via a bridge equation, the nature of the microscopic degrees of freedom is irrelevant for thermodynamic behaviour. In kinetic theory, we instead work with the microscopic degrees of freedom directly to calculate macroscopic quantities. Kinetic theory also gives a means to calculate transport properties, such as conductivities or diffusion coefficients. We first apply kinetic theory to the familiar system of a classical ideal gas, which allows us to view it from a different point of view to the one we have taken thus far, that it is the classical limit of N non-interacting particles confined in a box. To do this we will derive the distribution of particle velocities in an ideal gas, the Maxwell–Boltzmann distribution. We then use a simple application of kinetic theory to obtain the average pressure and the effusion rate for an ideal gas. We finish with a discussion of Brownian motion and diffusion.

5.1 Maxwell–Boltzmann Velocity Distribution

We know from the canonical ensemble that for an ideal gas the probability that the system is in any particular microstate with energy between E and $E + dE$ and partition function Z is given by

$$\frac{e^{-\beta E}}{Z}.$$

As we discussed in [Section 4.5.3](#), we can understand the properties of an ideal gas by considering a single particle and then obtain the partition function of a many-particle system from the single-particle partition function [Eq. \(4.85\)](#):

$$Z_1 = g_s n_Q V = g_s V \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3}{2}}. \quad (5.1)$$

In order to find the probability that the system has energy E , we need to know how many microstates there are that have energy within the energy range $[E, E + dE]$, which we calculate below.

5.1.1 Density of States

To obtain the density of states, we first pick an energy E^* and then calculate the number of states with energy $E < E^*$. For particle in a box states in three dimensions, for a cubic box with side length L and volume $V = L^3$, we can write the energy eigenvalues as

$$E = \frac{\hbar^2}{2m} \frac{\pi^2}{V^{2/3}} (n_x^2 + n_y^2 + n_z^2), \quad (5.2)$$

where n_x , n_y and n_z are positive integers, so we require

$$n_x^2 + n_y^2 + n_z^2 < \left(\frac{2mE^*}{\hbar^2} \right) \frac{V^{2/3}}{\pi^2}, \quad (5.3)$$

which represents 1/8 of a sphere (in energy space) with radius $r = \sqrt{\frac{2mE^*}{\hbar^2}} \frac{V^{1/3}}{\pi}$. Now, for a sphere of radius r , the number of points (n_x, n_y, n_z) in the octant of the sphere, where $n_x, n_y, n_z > 0$, is $\frac{1}{8} \times \frac{4}{3} \pi r^3 = \frac{1}{6} \pi r^3$, so the number of states in the relevant portion of the sphere with energy $\leq E$ is (including a factor g_s to take into account spin degeneracy) is

$$N(E) = \frac{g_s \pi}{6} \left(\frac{2mE}{\hbar^2} \right)^{3/2} \frac{V}{\pi^3} = \frac{g_s}{6\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V E^{3/2}. \quad (5.4)$$

If we look in the energy range $[E, E + dE]$, then the number of states is

$$\begin{aligned} N(E + dE) - N(E) &\simeq \frac{dN}{dE} dE \\ &= \frac{g_s}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V E^{1/2} dE. \end{aligned} \quad (5.5)$$

Thus the density of states, i.e. the number of states per unit energy (which we can also view as the degeneracy of states at energy E) is

$$g(E) = \frac{dN}{dE} = \frac{g_s}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} V E^{1/2}. \quad (5.6)$$

We discuss an alternative method to calculate the density of states in [Chapter 7](#).

5.1.2 Maxwell–Boltzmann Velocity Distribution

Armed with the density of states, we can calculate the probability density that the system has energy E by noting that the probability that the energy lies between E and $E + dE$ is

$$\begin{aligned}
p(E)dE &= g(E) \frac{e^{-\beta E}}{Z} dE \\
&= \frac{g_s}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} V E^{\frac{1}{2}} \frac{1}{g_s V \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{\frac{3}{2}}} e^{-\beta E} dE \\
&= \frac{2}{\pi^2} \left(\frac{\pi}{k_B T} \right)^{\frac{3}{2}} E^{\frac{1}{2}} e^{-\beta E} dE.
\end{aligned} \tag{5.7}$$

We can move from a probability density in energy to a probability density in velocity (or equivalently momentum $\mathbf{p} = m\mathbf{v}$) using $E = \frac{1}{2}mv^2$, where the speed v is the magnitude of the velocity $\mathbf{v}$, which implies $dE = mv dv$. This gives

$$p(v)dv = \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} e^{-\frac{mv^2}{2k_B T}} 4\pi v^2 dv. \tag{5.8}$$

The probability density we have found in Eq. (5.8) is implicitly one in phase space, $P(\mathbf{x}, \mathbf{p})$, but since the energy of a gas molecule is independent of its position and its direction of motion, we can write

$$\int d^3\mathbf{x} \int d^3\mathbf{p} P(\mathbf{x}, \mathbf{p}) = \int_0^\infty dv p(v) = 1. \tag{5.9}$$

From Eq. (5.9), if we treat all the particles in the gas as equivalent, we can infer that the number density of particles that have velocities within $d^3\mathbf{v} = dv_x dv_y dv_z$ of $\mathbf{v}$, which we will denote by $f(\mathbf{v})d^3\mathbf{v}$, satisfies

$$n_0 = \frac{N}{V} = \frac{N}{V} \int d^3\mathbf{x} \int d^3\mathbf{p} P(\mathbf{x}, \mathbf{p}) = n_0 \int_0^\infty dv p(v) = \int d^3\mathbf{v} f(\mathbf{v}), \tag{5.10}$$

where n_0 is the number density of the particles. By rewriting the integral on the right-hand side and noting that there is no preferred direction in space (so $f(\mathbf{v})$ should be a function of v only), we see that

$$n_0 \int_0^\infty dv p(v) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin\theta \int_0^\infty v^2 dv f(\mathbf{v}) = \int_0^\infty 4\pi v^2 dv f(\mathbf{v}), \tag{5.11}$$

which allows us to read off the **Maxwell–Boltzmann velocity distribution** for particle velocities in an ideal gas:

$$f(\mathbf{v}) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} e^{-\frac{mv^2}{2k_B T}}. \tag{5.12}$$

To obtain this result, we used the fact that in an ideal gas there are no interactions between particles, so the distribution of values of velocity for a single particle will mirror the distribution of values of velocity for a gas with many particles.

5.2 Properties of the Maxwell–Boltzmann Velocity Distribution

We now identify some important features of the Maxwell–Boltzmann velocity distribution. The first is the normalization such that

$$\int d^3\mathbf{v} f(\mathbf{v}) = n_0, \quad (5.13)$$

where n_0 is the number density of the gas. In our derivation we assumed that there is no flow, so that we are in the rest frame of the fluid and $\langle \mathbf{v} \rangle = 0$.¹ If there is a net flow, so that $\langle \mathbf{v} \rangle \neq 0$, then we can shift from the rest frame of the fluid, in which case

$$f(\mathbf{v}) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} e^{-\frac{m(\mathbf{v}-\langle \mathbf{v} \rangle)^2}{2k_B T}}. \quad (5.14)$$

We set $\langle \mathbf{v} \rangle = 0$ in our following discussion.

Recalling that $v^2 = v_x^2 + v_y^2 + v_z^2$, we see that the Maxwell–Boltzmann distribution factorizes into three identical distributions for v_x , v_y , v_z , i.e.

$$f(\mathbf{v}) = n_0 g(v_x) g(v_y) g(v_z), \quad (5.15)$$

where

$$g(v_x) dv_x = \frac{1}{n_0} dv_x \int \int dv_y dv_z f(\mathbf{v}) = \left(\frac{m}{2\pi k_B T} \right)^{\frac{1}{2}} e^{-\frac{mv_x^2}{2k_B T}} dv_x, \quad (5.16)$$

and $n_0 g(v_x) dv_x$ is the density of particles with velocities in the x direction between v_x and $v_x + dv_x$. We see that $g(v_x)$ is a Gaussian distribution peaked at $v_x = 0$; $g(v_y)$ and $g(v_z)$ have similar forms. This observation can be helpful for calculating moments of the velocity distribution

It is helpful to calculate the first two moments of the distribution, which correspond to the average speed and mean square velocity, respectively (when $\langle \mathbf{v} \rangle = 0$). The first moment is

$$\begin{aligned} \langle v \rangle &= \frac{\int d^3\mathbf{v} v f(\mathbf{v})}{\int d^3\mathbf{v} f(\mathbf{v})} \\ &= \frac{1}{n_0} n_0 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} 4\pi \int_0^\infty dv v^3 e^{-\frac{mv^2}{2k_B T}} \\ &= \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} 4\pi \left(\frac{2k_B T}{m} \right)^2 \int_0^\infty dx x^3 e^{-x^2} \\ &= \sqrt{\frac{8k_B T}{\pi m}} \\ &\simeq 1.60 \sqrt{\frac{k_B T}{m}}, \end{aligned} \quad (5.17)$$

¹ A more sophisticated derivation allowing for flow follows from solving the Boltzmann equation allowing for collisions between gas molecules.

where we let $x^2 = mv^2/2k_B T$ in going from the second to the third line to obtain Eq. (5.17). The second moment can be calculated as

$$\langle v^2 \rangle = \frac{3k_B T}{m}, \quad (5.18)$$

either by direct integration, or using the equipartition theorem. This corresponds to a root mean square velocity of

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}} \simeq 1.73 \sqrt{\frac{k_B T}{m}}. \quad (5.19)$$

5.2.1 Distribution of Speeds

We can also determine a distribution of speeds (rather than velocities) by rewriting the Maxwell–Boltzmann velocity distribution, i.e.

$$\begin{aligned} F(v)dv &= \int_0^{2\pi} d\phi \int_0^\pi \sin\theta d\theta f(\mathbf{v}) v^2 dv \\ &= 4\pi v^2 f(\mathbf{v}) dv \\ &= 4\pi n_0 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2k_B T}} dv. \end{aligned} \quad (5.20)$$

A third velocity scale that enters naturally when we look at the distribution of speeds is the most probable speed v_p , which corresponds to the maximum in $F(v)$. This maximum is given by

$$\frac{dF}{dv} = 0, \quad (5.21)$$

which implies

$$\left(2v - \frac{mv^3}{k_B T} \right) e^{-\frac{mv^2}{2k_B T}} = 0 \quad (5.22)$$

when $v = v_p$, i.e.

$$v_p = \sqrt{\frac{2k_B T}{m}} \simeq 1.41 \sqrt{\frac{k_B T}{m}}. \quad (5.23)$$

We note that v_p , $\langle v \rangle$ and v_{rms} are all very similar in magnitude, but are ordered as $v_p < \langle v \rangle < v_{\text{rms}}$. Which one is relevant in a given physical situation depends on the question being asked. These speeds and the entire distribution are illustrated in Fig. 5.1.

Example: Speed of Molecules in Air

We can use the results we have obtained for $\langle v \rangle$, v_{rms} and v_p to calculate the speed of molecules in air. We approximate air as 78% nitrogen, 21% oxygen and 1% argon,² which

² The composition of dry air is more precisely given by 78.09% nitrogen, 20.95% oxygen, 0.93% argon and 0.04% carbon dioxide, with trace amounts of other gases. Water vapour is also present in the atmosphere; at sea level this is usually around the 1% level.

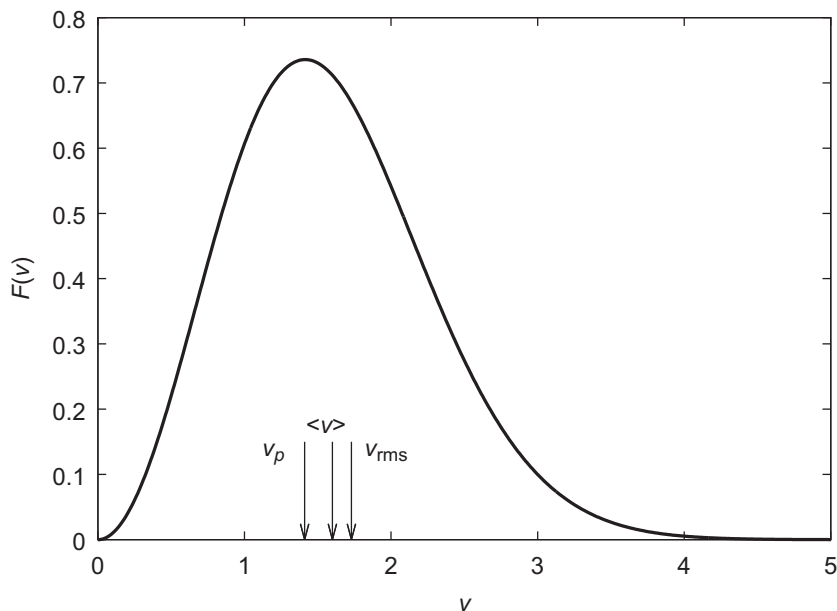


Fig. 5.1

Distribution of speeds that follows from the Maxwell–Boltzmann velocity distribution with positions of the average velocity $\langle v \rangle$, most probable velocity v_p and root mean square velocity v_{rms} , marked. Velocity is measured in units of $\sqrt{\frac{k_B T}{m}}$ and the distribution is not normalized.

have molecular masses of 28.01 amu (nitrogen), 32.00 amu (oxygen) and 39.95 amu (argon). The average mass of a molecule in air is thus 29.0 amu, so we can calculate for air at a temperature of 300 K that

$$\sqrt{\frac{k_B T}{m}} \simeq \sqrt{\frac{1.3807 \times 10^{-23} \times 300}{29.0 \times 1.66 \times 10^{-27}}} = 293 \text{ m s}^{-1}. \quad (5.24)$$

Thus we get the following measures for the speed of molecules in air:

$$\begin{aligned} \langle v \rangle &\simeq 1.60 \sqrt{\frac{k_B T}{m}} \sim 470 \text{ m s}^{-1}, \\ v_{\text{rms}} &\simeq 1.73 \sqrt{\frac{k_B T}{m}} \sim 510 \text{ m s}^{-1}, \\ v_p &\simeq 1.41 \sqrt{\frac{k_B T}{m}} \sim 410 \text{ m s}^{-1}, \end{aligned}$$

indicating that characteristic speeds of molecules in air are in the range 400–500 m s^{−1}.

5.3 Kinetic Theory for an Ideal Gas

We now make use of kinetic theory to calculate the properties of a dilute gas in a finite container.

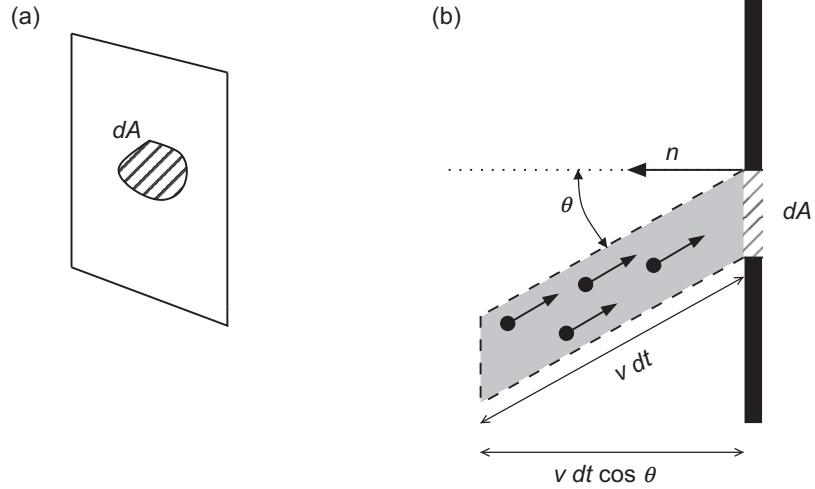


Fig. 5.2

(a) Schematic of area dA on wall of container; (b) cylinder containing all molecules with a velocity directed at an angle θ to the wall with speed between v and $v + dv$ that are available to collide with the element of area dA in time dt .

5.3.1 Pressure in an Ideal Gas

Molecules in a gas collide with the walls of their container and exchange momentum with the walls, which leads to a force per unit area that can be identified as the pressure. We can calculate the pressure by determining the momentum transfer to the wall. To do this, we consider a fluid at rest and focus on an area dA on the wall, as depicted in Fig. 5.2(a). If we consider a short time interval dt , we can calculate how many molecules are incident on dA with velocities between $\mathbf{v}$ and $\mathbf{v} + d\mathbf{v}$. This will be exactly the number of molecules in the cylinder shown in Fig. 5.2(b), i.e.

$$\underbrace{f(\mathbf{v}) d^3\mathbf{v}}_{\text{Density of molecules}} \times \underbrace{v_n dA dt}_{\text{Volume of cylinder}},$$

where $v_n = \mathbf{v} \cdot \hat{\mathbf{n}}$ with $\hat{\mathbf{n}}$ the unit normal to the surface. We can also write $v_n = v \cos \theta$ in the co-ordinate system used in Fig. 5.2(b). To determine the total number of molecules colliding with dA , we need to integrate over all velocities with positive v_n (molecules with negative v_n move away from the surface).

Similarly, we can determine the momentum carried to the wall by the molecules in the cylinder as

$$\underbrace{(m\mathbf{v}) f(\mathbf{v}) d^3\mathbf{v}}_{\text{Momentum density}} \times \underbrace{v_n dA dt}_{\text{Volume of cylinder}},$$

hence the total incident momentum on the region dA is

$$dA dt \int_+ d^3\mathbf{v} m\mathbf{v} f(\mathbf{v}) v_n,$$

where the + subscript on the integral sign means that $v_n > 0$, and the total momentum reflected by the surface is

$$dA dt \int_{-} d^3\mathbf{v} m\mathbf{v} f(\mathbf{v}) (-v_n).$$

These two momenta point in opposite directions, so there is momentum $2mv_n$ transferred in each collision. The total momentum $d\mathbf{p}$ exchanged between the gas and the wall in time dt via the surface dA is thus

$$\begin{aligned} d\mathbf{p} &= dA dt \left[\int_{+} d^3\mathbf{v} m\mathbf{v} f(\mathbf{v}) v_n - \int_{-} d^3\mathbf{v} m\mathbf{v} f(\mathbf{v}) (-v_n) \right] \\ &= dA dt \int d^3\mathbf{v} m\mathbf{v} f(\mathbf{v}) v_n \\ &= dA dt n_0 m \langle v_n \mathbf{v} \rangle. \end{aligned} \quad (5.25)$$

Identifying $d\mathbf{p}/dt$ as the force applied to the wall by molecular collisions, and dividing by dA , we have a force per unit area, which allows us to determine the force $\mathbf{F}$ per area A on the surface with normal $\hat{\mathbf{n}}$ as

$$\begin{aligned} \frac{\mathbf{F}}{A} &= n_0 m \langle v_n \mathbf{v} \rangle \\ &= \rho \sum_{i=1}^3 \hat{n}_i \langle v_i \mathbf{v} \rangle, \end{aligned} \quad (5.26)$$

where $\rho = n_0 m$ is the mass density of the gas. We notice that in the expression Eq. (5.26), $\langle v_n \mathbf{v} \rangle = 0$ for any of the components of $\mathbf{v}$ that are orthogonal to $\hat{\mathbf{n}}$ for an ideal gas.³

The force per unit area exerted on the wall by the fluid is the stress, Σ , which we can write as (note the minus sign, which arises because the force is directed oppositely to the normal to the surface)

$$\Sigma_i = -\frac{F_i}{A} = \sum_{j=1}^3 \sigma_{ij} \hat{n}_j, \quad (5.27)$$

where σ_{ij} is the stress tensor. We can read off from Eqs (5.26) and (5.27) that⁴

$$\sigma_{ij} = -\rho \langle v_i v_j \rangle. \quad (5.28)$$

³ To illustrate this point, suppose that $\hat{\mathbf{n}} = \hat{\mathbf{z}}$, in which case $v_n = v_z$, and then

$$\langle v_z \mathbf{v} \rangle = \begin{bmatrix} \langle v_z v_x \rangle \\ \langle v_z v_y \rangle \\ \langle v_z^2 \rangle \end{bmatrix} = \frac{1}{n_0} \int_{-\infty}^{\infty} dv_x \int_{-\infty}^{\infty} dv_y \int_{-\infty}^{\infty} dv_z \begin{bmatrix} v_z v_x \\ v_z v_y \\ v_z^2 \end{bmatrix} f(\mathbf{v}),$$

and since $f(\mathbf{v})$ is an even function of v_x , v_y , and v_z , we see that $\langle v_z v_x \rangle = \langle v_z v_y \rangle = 0$ and only $\langle v_z^2 \rangle \neq 0$.

⁴ Recall that we assumed a fluid at rest. If a fluid is in motion with an average velocity of $\langle \mathbf{v} \rangle = \mathbf{u}$, then we can write the velocity of molecules in the fluid as $\mathbf{v} = \mathbf{u} + \mathbf{w}$, where $\mathbf{w}$ describes the random motion of molecules in the fluid and $\langle \mathbf{w} \rangle = 0$. The stress tensor is defined to only include the random motion of the molecules and not any overall flow, and hence in general takes the form

$$\sigma_{ij} = -\rho \langle w_i w_j \rangle.$$

Our discussion indicates that the stress tensor is (in principle) not necessarily isotropic and can be represented by the 3×3 matrix σ_{ij} , which is a second-rank tensor. The component σ_{ij} is the force per unit area in the j direction exerted on a surface with normal in the i direction. The off-diagonal elements correspond to a shear stress (forces applied parallel to the surface) and the diagonal elements to a tensile stress (forces applied normal to the surface).

A gas cannot support shear stresses, so the off-diagonal components of the stress tensor vanish (which can also be checked by explicit calculation for the Maxwell–Boltzmann velocity distribution). Hence

$$\sigma_{ii} = -\rho \langle v_i^2 \rangle, \quad (5.29)$$

and $\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle = \frac{1}{3} \langle v^2 \rangle$ due to the isotropy of the Maxwell–Boltzmann velocity distribution, as can be verified by direct computation. Thus the stress is isotropic and has mean value

$$\sigma = -\frac{1}{3} n_0 m \langle v^2 \rangle = -\frac{1}{3} n_0 m \frac{3k_B T}{m} = -n_0 k_B T = -\frac{N k_B T}{V}, \quad (5.30)$$

which corresponds to a force per unit area exerted on the surface (i.e. a pressure)

$$P = \frac{F}{A} = \frac{N k_B T}{V}, \quad (5.31)$$

as we expect for an ideal gas. Provided we consider the special case of isotropic fluids (i.e. fluids which have the same properties in all directions), such as an ideal gas, we only need to worry about the pressure and none of the other components of the stress tensor. However, when dealing with complex fluids which may be anisotropic, such as liquid crystals, the tensor nature of stress becomes important.

5.3.2 Effusion

In our calculation of the pressure, we calculated the number of particles striking the area dA in time dt for a given velocity $\mathbf{v}$. Integrating over all velocities (subject to the restriction that $v_n > 0$), the total number of particles striking the area dA is

$$N_{\text{incident}} = dA dt \int_+ d^3\mathbf{v} f(\mathbf{v}) v_n. \quad (5.32)$$

In effusion we consider the situation when the area dA is removed, so that there is a small hole in the side of the container and gas molecules can escape from the container (Fig. 5.3). In that case, the value of N_{incident} in Eq. (5.32) is the total number of particles escaping the container through the hole in time dt . Hence we can calculate the flux of molecules escaping through the hole per unit area per unit time as

$$\begin{aligned} \int_+ d^3\mathbf{v} f(\mathbf{v}) v_n &= \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta \sin \theta \int_0^\infty v^2 dv v \cos \theta f(\mathbf{v}) \\ &= 2\pi \int_0^1 dx x \int_0^\infty v^2 dv v f(\mathbf{v}) \end{aligned}$$

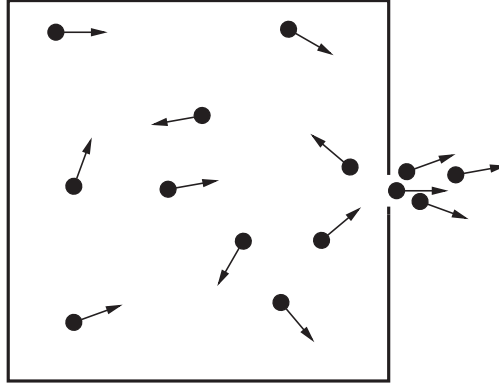


Fig. 5.3

Illustration of effusion – particles can escape from a small hole in a container.

$$\begin{aligned}
 &= 2\pi \left[\frac{1}{2} x^2 \right]_0^1 \int_0^\infty v^2 dv v f(\mathbf{v}) \\
 &= \pi \int_0^\infty v^2 dv v f(\mathbf{v}) \\
 &= \frac{n_0}{4} \langle v \rangle,
 \end{aligned} \tag{5.33}$$

where we used the substitution $x = \cos \theta$ and recalled that the Maxwell–Boltzmann velocity distribution is isotropic in $\mathbf{v}$, and that the angular integral which appears in the expectation value $\langle v \rangle$ is

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta = 4\pi. \tag{5.34}$$

One application of effusion is in the enrichment of uranium. The rate of effusion we calculated above is

$$\frac{n_0}{4} \langle v \rangle = \frac{1}{4} n_0 \sqrt{\frac{8k_B T}{\pi m}} \simeq 0.4 n_0 \sqrt{\frac{k_B T}{m}}, \tag{5.35}$$

which depends on the mass of the molecules. Uranium hexafluoride gas molecules will have slightly different masses depending on whether they contain the ^{235}U isotope or the ^{238}U isotope, and hence slightly different rates of effusion. By repeated application of effusion it is possible to increase the proportion of the isotope ^{235}U which is used for fission in either nuclear reactors or nuclear weapons.

5.4 Brownian Motion

During Brownian motion a particle in a fluid moves in a seemingly random way, as was observed for pollen grains in water by Robert Brown. This motion comes from the interaction of the particle with the smaller molecules in the fluid. We will

illustrate Brownian motion using a one-dimensional example for simplicity. Let the position of the particle of interest, e.g. a pollen grain, be x , in which case its velocity is given by $v = dx/dt$. The total force experienced by the particle will have two contributions: (i) an external force, $\mathcal{F}(t)$ and (ii) a net force, $F(t)$ that arises from the interactions of the particles with the molecules in the fluid.⁵ Newton's second law gives

$$m \frac{dv}{dt} = \mathcal{F}(t) + F(t). \quad (5.36)$$

Now, $F(t)$ arises from the collisions of molecules in the fluid with the particle of interest. These molecules are moving randomly at very fast speeds and hence $F(t)$ varies rapidly in time in both sign and magnitude. If we are interested in the average motion of the particle, rather than for a very short period of time, then we can average over $F(t)$. Suppose that τ_0 is the time scale over which $F(t)$ fluctuates, then for times $T \gg \tau_0$, there is no net force from fluctuations, i.e.

$$\langle F \rangle = \frac{1}{T} \int_0^T d\tilde{t} F(\tilde{t}) = 0. \quad (5.37)$$

To take advantage of this observation about the nature of $F(t)$, write the velocity as $v = u + \delta v$, where δv fluctuates on time scales of order τ_0 so that the time average $\langle \delta v \rangle = 0$ and hence u gives the slower average motion on time scales $\gg \tau_0$.

We expect that the force on the particle from the fluid should act so as to bring the particle to equilibrium, so it will depend on the particle velocity, in which case we can write the fluctuating force in the form

$$F(t) = F_u(t) + \delta F(t), \quad (5.38)$$

where $\langle \delta F(t) \rangle = 0$ and $F_u(t)$ is a slowly varying part of the force, that we can expand in powers of u . When $u = 0$, we expect that $F_{u=0}(t) = 0$, and so the lowest term in the expansion will be the linear term. Hence we may approximate

$$F_u(t) \simeq -\alpha u, \quad (5.39)$$

where α is a constant that quantifies the damping of motion of the particle by the fluid. This gives us the following equation for the slowly varying part of the velocity:

$$m \frac{du}{dt} = \mathcal{F}(t) - \alpha u. \quad (5.40)$$

We can thus rewrite Eq. (5.36) to obtain the **Langevin equation** (Langevin, 1908)

$$m \frac{dv}{dt} = \mathcal{F}(t) - \alpha v + \delta F(t), \quad (5.41)$$

⁵ The quantitative link between the idea that molecules in the fluid exert random forces on a particle was provided by Einstein in 1905, the same year that he also published his famous papers on special relativity and the photoelectric effect.

where we assumed that $\alpha u \simeq \alpha v$ since the error we have introduced, $\alpha \delta v$, should be negligible in comparison to $\delta F(t)$.

We first consider the Langevin equation in the absence of an external force $\mathcal{F}(t)$. It is instructive to consider the situation when there is no random force, $\delta F(t)$, in which case we have

$$m \frac{dv}{dt} = -\alpha v, \quad (5.42)$$

which has the solution

$$v(t) = v_0 e^{-\frac{t}{\tau}}, \quad (5.43)$$

where we defined $\tau = m/\alpha$. In this case, the velocity drops to zero exponentially and the solution of the equation does not accord with the phenomenology of Brownian motion.

We now reintroduce the rapidly fluctuating force term. Recall that $v = \dot{x} = dx/dt$ and multiply both sides of Eq. (5.41) by x , in which case we get

$$mx \frac{d\dot{x}}{dt} = m \frac{d(x\dot{x})}{dt} - m\dot{x}^2 = -\alpha x\dot{x} + x\delta F(t). \quad (5.44)$$

Now, if we take an average over times $\gg \tau_0$, this becomes

$$m \left(\frac{d}{dt} + \alpha \right) \langle x\dot{x} \rangle = m \langle \dot{x}^2 \rangle + \langle x\delta F(t) \rangle, \quad (5.45)$$

and since the fluctuating force is uncorrelated with position, and also has vanishing mean (i.e. $\langle \delta F(t) \rangle = 0$), we have $\langle x\delta F(t) \rangle = \langle x \rangle \langle \delta F(t) \rangle = 0$. We can also use the equipartition theorem to evaluate $m \langle \dot{x}^2 \rangle$, which is twice the kinetic energy, and hence $m \langle \dot{x}^2 \rangle = k_B T$.⁶

Thus we can simplify Eq. (5.45) to

$$\left(\frac{d}{dt} + \frac{\alpha}{m} \right) \langle x\dot{x} \rangle = \frac{k_B T}{m}, \quad (5.46)$$

which has the solution (as can be verified by substitution)

$$\langle x\dot{x} \rangle = \frac{k_B T}{\alpha} \left(1 - e^{-\frac{t}{\tau}} \right), \quad (5.47)$$

where $\tau = m/\alpha$ takes the same value as in the absence of a random force and is the characteristic time scale for velocity relaxation for the particle. If we note that

$$\begin{aligned} \frac{d}{dt} \langle x^2(t) \rangle &= 2 \langle x\dot{x} \rangle \\ &= \frac{2k_B T}{\alpha} \left(1 - e^{-\frac{t}{\tau}} \right), \end{aligned} \quad (5.48)$$

then we can integrate to get

$$\langle x^2(t) \rangle = \frac{2k_B T}{\alpha} \left[t + \tau e^{-\frac{t}{\tau}} + C \right], \quad (5.49)$$

⁶ It might seem that we reintroduced the fluctuating force term only for it to drop out of the equation again when we took a temporal average. However, the presence of fluctuating forces allows for equilibrium to be established, which then allows us to use the equipartition theorem.

where C is a constant of integration. Requiring that $\langle x^2(t=0) \rangle = 0$ implies that $C = -\tau$, and hence

$$\langle x^2(t) \rangle = \frac{2k_B T}{\alpha} \left[t - \tau \left(1 - e^{-\frac{t}{\tau}} \right) \right]. \quad (5.50)$$

We can explore the behaviour of this expression in both the short and long time limits. For very short times, $t \ll \tau$, we can Taylor expand the exponential in Eq. (5.50) as

$$e^{-\frac{t}{\tau}} \simeq 1 - \frac{t}{\tau} + \frac{1}{2} \frac{t^2}{\tau^2} + \dots, \quad (5.51)$$

and obtain

$$\langle x^2(t) \rangle \simeq \frac{k_B T}{\alpha \tau} t^2, \quad (5.52)$$

while in the long time limit $t \gg \tau$, we obtain

$$\langle x^2(t) \rangle \simeq \frac{2k_B T}{\alpha} t. \quad (5.53)$$

The solutions to the Langevin equation we found in Eqs (5.52) and (5.53) mirror the behaviour we found for the random walk in Section 1.2.4, that at short times the rms displacement grows ballistically whereas for longer times it grows diffusively. If we record the position of a Brownian particle then its displacement will have the form of a random walk, as illustrated for one dimension in Fig. 1.4.

5.5 Diffusion

In our discussion of random walks in Chapter 1 (and in Problem 1.4) we saw that the continuum limit of a random walk corresponds to diffusion, in which a probability distribution in space and time, $\rho(\mathbf{x}, t)$, evolves according to the diffusion equation

$$\frac{\partial \rho}{\partial t} = D \nabla^2 \rho, \quad (5.54)$$

where D is the diffusion coefficient. The form of diffusion equation in Eq. (5.54), with one time and two space derivatives, hints at the relationship between characteristic time (t_c) and length (L_c) scales in diffusion processes, that is $L_c \sim \sqrt{t_c}$.

5.5.1 Mean Free Path and Collision Time

We can use kinetic theory to get insight into the microscopic origin of the diffusion coefficient D . To do this we will consider collisions, which correspond to when particles get close enough to each other to interact strongly. Rather than trying to describe every collision precisely, we treat them statistically, and characterize collisions with a few parameters. The first of these is the mean distance between collisions, the mean free path, which we denote by l .

In order to visualize the mean free path, treat the molecules in a gas as spheres of diameter d . There will be a collision if the centres of two spheres pass within a distance d

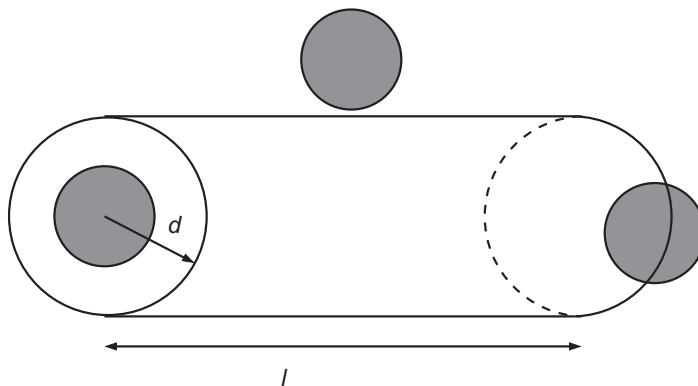


Fig. 5.4

Cylinder swept out between collisions for spherical molecules of diameter d in a gas with mean free path l .

of each other. In between collisions, there are no forces acting on the molecules, so they will travel in straight lines and sweep out a cylindrical volume of $l\pi d^2$, as illustrated in Fig. 5.4.⁷

On average, there should be a collision every time this cylindrical volume is equal to the volume per atom, $1/n$, where n is the number density, so

$$l\pi d^2 = \frac{1}{n}, \quad (5.55)$$

and so the mean free path is

$$l = \frac{1}{n\pi d^2}. \quad (5.56)$$

Having obtained a distance scale for collisions, we can also determine a time scale. For molecules moving with speed v , the collision time τ will be related to the mean free path by

$$l = v\tau. \quad (5.57)$$

If we take v to be the rms speed v_{rms} , introduced in Eq. (5.19), and use the expression we obtained above for the mean free path, Eq. (5.56), then we get an estimate⁸

$$\tau \simeq \frac{l}{v_{\text{rms}}} = \sqrt{\frac{m}{3\pi^2 k_B T}} \frac{1}{nd^2}. \quad (5.58)$$

Hence we see that the collision time scales as $\tau \sim T^{-\frac{1}{2}}$. This is in accord with our intuition that at higher temperatures molecules will move more quickly and have

⁷ Strictly, if we imagine the molecules interacting via a potential similar to the Lennard–Jones potential introduced in Eq. (4.111), the collisions correspond to when molecules are close enough to experience the strongly repulsive part of the potential, and in between collisions there are very weak attractions between the molecules. To a first approximation we can ignore the weak attractions that will cause molecules to deviate from straight-line motion and assume that only the repulsive forces are important, in which case the molecules move in straight lines between collisions.

⁸ Our first instinct might be that we should take v to be the average speed $\langle v \rangle$, but because the molecules are moving relative to each other, we should have the average relative speed, which a careful analysis reveals to be v_{rms} .

more collisions per unit time (and hence less time between collisions) than at low temperatures.

Example: Mean Free Path and Collision Time in Air

We can use Eqs (5.56) and (5.58) to estimate the mean free path and collision time for molecules in air. To calculate the mean free path, we need the number density n and the diameter d for molecules in air. We can obtain the number density at atmospheric pressure (1.013×10^5 Pa) and a temperature of 300 K from the ideal gas law as

$$n = \frac{P}{k_B T} = \frac{1.013 \times 10^5}{1.3807 \times 10^{-23} \times 300} \simeq 2.4 \times 10^{25} \text{ m}^{-3}.$$

We can also note that the average bond length in N_2 is $\sim 1.1 \text{ \AA}$ and the average bond length in O_2 is $\sim 1.2 \text{ \AA}$, so if we treat air to be 80% nitrogen and 20% oxygen, this gives $d \simeq 1.1 \text{ \AA}$. Hence the mean free path in air is

$$l = \frac{1}{n\pi d^2} = \frac{1}{2.4 \times 10^{25} \times \pi \times (1.1 \times 10^{-10})^2} \simeq 1 \text{ \mu m}.$$

Once we have the mean free path we can use our earlier estimate for the speed of molecules in air of $v_{\text{rms}} = 510 \text{ m s}^{-1}$ to get a collision time of

$$\tau \simeq \frac{l}{v_{\text{rms}}} \simeq \frac{10^{-6}}{510} \simeq 2 \text{ ns}.$$

5.5.2 Fick's Law

Consider a situation in which the average density of molecules is constant, but that some subset of the molecules is labelled, e.g. by being in a long-lived excited nuclear state, and this subset of molecules is not uniformly distributed, so that their number density, n , is position dependent. Further, simplify to the case in which the non-uniformity is only in the z direction. Collisions between the labelled and unlabelled molecules will tend to mix the two components so that after a sufficiently long time the labelled molecules are distributed uniformly in space. We can investigate the relaxation from the original out-of-equilibrium initial condition towards a uniform equilibrium state via diffusion.

If we consider a plane at constant z we can investigate the flux of labelled particles across that plane, J_z , which will be the mean number of labelled particles crossing the plane per unit area per unit time. We can apply the ideas of kinetic theory that we have developed in this chapter to calculate this flux in a gas by using a similar approach to Fig. 5.2, as illustrated in Fig. 5.5. If we let the number density of particles at a given z be $n(z)$, then we can calculate the flux coming up through the plane at fixed z for a given θ as

$$J_z = \frac{dN}{dA dt} = v_z \times \underbrace{\frac{n(z - l \cos \theta)}{n_0}}_{\text{Number of molecules}} f(\mathbf{v}) d^3 \mathbf{v}, \quad (5.59)$$

where we have assumed that particles travel a mean free path before crossing the plane (the factor of $\cos \theta$ is because the particle velocities are not guaranteed to be normal to the

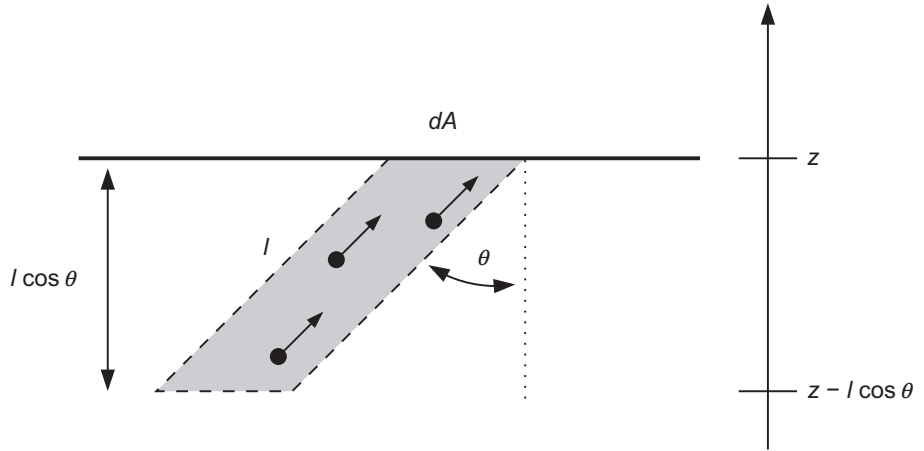


Fig. 5.5

Geometry used for obtaining Fick's law. Particles are assumed to travel a mean free path l before crossing the plane at z with an incoming angle of θ .

surface and hence the change in z for a particle travelling a mean free path and having a velocity oriented at angle θ to the plane is $l \cos \theta$). In practice, not all molecules will have travelled as far as a mean free path before reaching our imaginary surface and so, while our estimate of the diffusion coefficient should capture the correct qualitative behaviour, a more sophisticated treatment would give a more quantitatively accurate result. We can perform a similar estimate to Eq. (5.59) for the flux coming from above the plane and when we sum the two contributions we get Fick's law:

$$\begin{aligned}
 J_z &= \frac{1}{n_0} \int d^3\mathbf{v} v \cos \theta [n(z - l \cos \theta) - n(z + l \cos \theta)] f(\mathbf{v}) \\
 &\simeq -\frac{2}{n_0} \left\{ \int d^3\mathbf{v} v l \cos^2 \theta f(\mathbf{v}) \right\} \frac{\partial n}{\partial z} \\
 &= -D \frac{\partial n}{\partial z},
 \end{aligned} \tag{5.60}$$

where we expanded

$$n(z \pm l \cos \theta) \simeq n(z) \pm l \cos \theta \frac{\partial n}{\partial z}, \tag{5.61}$$

and combined all of the terms in front of the gradient into a constant:

$$\begin{aligned}
 D &= 2l \frac{\int_0^{\pi/2} d\theta \sin \theta \cos^2 \theta}{\int_0^{\pi} d\theta \sin \theta} \langle v \rangle \\
 &= 2l \frac{\left[-\frac{1}{3} \cos^3 \theta \right]_0^{\pi/2}}{\left[-\cos \theta \right]_0^{\pi}} \langle v \rangle \\
 &= \frac{1}{3} l \langle v \rangle.
 \end{aligned} \tag{5.62}$$

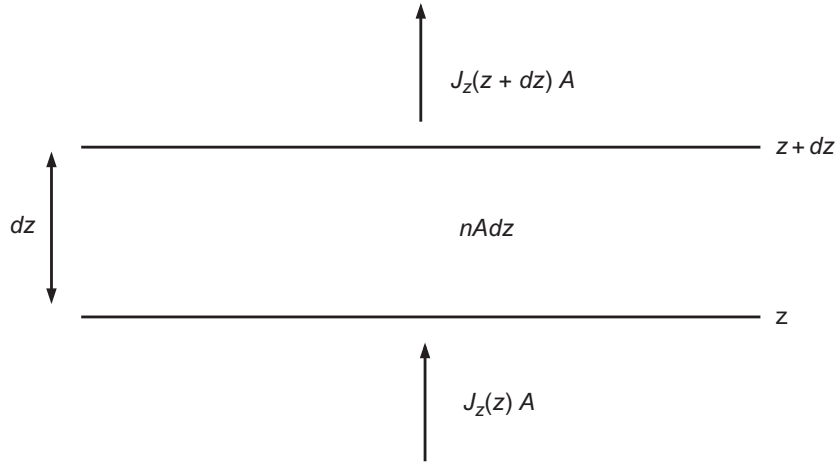


Fig. 5.6

Number of molecules in a slab of thickness dz and area A , $n(z)Adz$, and the flux of molecules entering $AJ_z(z)$ and leaving $AJ_z(z + dz)$ the slab.

The form of Fick's law, Eq. (5.60), implies that the concentration n satisfies the diffusion equation. To see this, consider a thin slab of thickness dz with lower surface at z and upper surface at $z + dz$, as illustrated in Fig. 5.6. The rate of change of the number of labelled particles in the slab, $nAdz$, will be given by the difference in particle numbers entering through the two surfaces per unit time, where we note that the rate at which particles enter or leave the slab will be given by the area times the flux, i.e.

$$\frac{\partial}{\partial t}(nAdz) = AJ_z(z) - AJ_z(z + dz), \quad (5.63)$$

which we can simplify to

$$\frac{\partial n}{\partial t} dz = -\frac{\partial J_z}{\partial z} dz, \quad (5.64)$$

and when we substitute in Eq. (5.60) for J_z , we obtain the diffusion equation for n :

$$\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial z^2}, \quad (5.65)$$

confirming that D is the diffusion coefficient.

5.6 Transport

Kinetic theory is particularly useful for calculating properties of systems that are perturbed so that there is a flow of some quantity across the system, which is known as transport. The example of diffusion in Section 5.5 is an example of transport: the particle flux J_z is related to the gradient of the particle density gradient by the diffusion coefficient D . Other examples of transport include the flow of energy, entropy or charge.

In the example of the flow of charge under the influence of an electric field, suppose that a potential difference V is applied across a distance L in a metal, giving rise to a uniform electric field $E = V/L$. Then the potential difference gives rise to a flow of charge, i.e. a current I . Provided the electric field is not too strong, the current will be related to the potential difference by

$$I = GV, \quad (5.66)$$

where G is the conductance. The inverse of the conductance is the resistance $R = 1/G$, so that Eq. (5.66) is equivalent to Ohm's law:

$$V = IR. \quad (5.67)$$

It is often more convenient to work with current density, $\mathbf{j} = I/A$, the current per unit area, with A the cross-sectional area of the metal, in which case we can write the relation as

$$\mathbf{j} = \sigma \mathbf{E}, \quad (5.68)$$

where $\sigma = G/AL$ is the conductivity, which has the advantage that it does not depend on the dimensions of the sample. This also has a similar form to Fick's law, Eq. (5.60), in that a current density ($\mathbf{j}$) is proportional to a gradient ($\mathbf{E} = -dV/dx$).

5.7 Summary

Kinetic theory gives a framework to calculate macroscopic physical properties of systems from the motion of their microscopic degrees of freedom. We can apply this to an ideal gas to find that the distribution of particle velocities in an ideal gas is given by the Maxwell–Boltzmann velocity distribution

$$f(\mathbf{v}) = n_0 \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}} e^{-\frac{m\mathbf{v}^2}{2k_B T}}. \quad (5.69)$$

The Maxwell–Boltzmann velocity distribution can be shown to reproduce the ideal gas law, and used to calculate the rate of effusion.

When collisions between molecules are important, diffusion can take place. For a small object that is still much larger than molecular length scales, such as a pollen grain, subjected to molecular collisions, Brownian motion can occur, which is described by the Langevin equation

$$m \frac{dv}{dt} = \mathcal{F}(t) - \alpha v + \delta F(t). \quad (5.70)$$

At the molecular scale, important parameters for describing collisions between molecules are the mean free path l and the collision time τ .

Problems

5.1 Calculate the following quantities for a three-dimensional classical ideal gas:

- (a) $\langle v_x \rangle$,
- (b) $\langle v_x^2 \rangle$,
- (c) $\langle v^2 v_x \rangle$,
- (d) $\langle v_x^3 v_y \rangle$,
- (e) $\langle (v_x + b v_y)^2 \rangle$ (b a constant),
- (f) $\langle v^2 v_x^2 \rangle$.

5.2 To cool atoms down to temperatures low enough to observe Bose–Einstein condensation (discussed in [Chapter 9](#)), experimentalists use the technique of evaporative cooling. The idea of the technique is to remove the most energetic particles, and then let the resulting distribution thermalize to a lower temperature than previously. We can illustrate this idea with a Maxwell–Boltzmann distribution. If we wish to remove the most energetic fraction α of the particles, then we can remove all particles with speed greater than some $\tilde{v}$. Show that

$$\alpha = \frac{4}{\pi^{\frac{1}{2}}} \int_{\tilde{x}}^{\infty} dx x^2 e^{-x^2},$$

and relate $\tilde{x}$ to $\tilde{v}$.

If the gas is allowed to thermalize after the most energetic particles are removed, then show that the relationship between the new and old temperatures, T_{new} and T_{old} , is given by

$$T_{\text{new}} = \frac{8T_{\text{old}}}{3\pi^{\frac{1}{2}}} \int_0^{\tilde{x}} dx x^4 e^{-x^2} < T_{\text{old}}.$$

Find the value of $\tilde{x}$ that gives $\alpha = 0.1$ and hence find T_{new} in terms of T_{old} (you will need to evaluate the integrals numerically).

5.3 Naturally occurring uranium primarily consists of the two isotopes ^{235}U (0.7% abundance) and ^{238}U (99.3% abundance). Only the lighter of the two isotopes is useful for nuclear fission. One method to increase the abundance of the lighter isotope is to use chemical processes to obtain the gas uranium hexafluoride UF_6 from natural uranium. This gas, consisting of 0.7% $^{235}\text{UF}_6$ and 99.3% $^{238}\text{UF}_6$, is then put into a container with a small hole, through which the gas can escape into vacuum, i.e. effusion.

- (a) Explain why the fraction of the gas escaping from the container is richer in $^{235}\text{UF}_6$ than the mixture left in the container.
- (b) Give an estimate of how much the abundance of $^{235}\text{UF}_6$ among the molecules escaping the container is increased compared to the initial abundance of 0.7%.

Note: the atomic mass of fluorine is 18.998 amu.

- 5.4** The pressure at the top of Mount Everest is about one-third of atmospheric pressure (101.3 kPa) and the temperature can be on the order of -30°C . Calculate the mean free path in air at this location and compare it to the result we found in [Section 5.5.1](#).
- 5.5** Giant molecular clouds are found in interstellar space and are composed mainly of hydrogen (H_2) molecules. They are a location where star formation can take place. A typical giant molecular cloud has a mass of $\sim 2 \times 10^5 M_\odot$, where $M_\odot \simeq 2 \times 10^{30} \text{ kg}$ is the mass of the Sun, a size of ~ 150 light years and an average temperature of 10 K. What is the mean free path and collision time scale for hydrogen molecules in such a cloud? (The H–H bond length is $0.74 \text{ \AA}$.)
- 5.6** Consider a fluid confined between two parallel plates, one with velocity $u_x = u_0$ and the other at rest ([Fig. 5.7](#)).

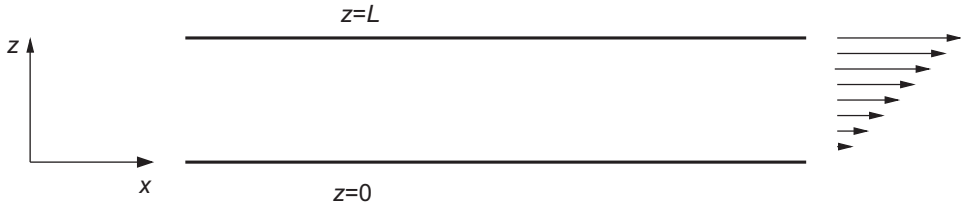


Fig. 5.7

Velocity gradient for fluid confined between two parallel plates.

The motion of the plates sets up a velocity gradient within the fluid as the fluid next to each plate takes the velocity of the respective plate. Layers with different velocities exert a force on each other, because there is a change in momentum for particles that move between layers with different velocities. Writing this as a force per unit area, which acts in the x direction and is exerted on a plane which is perpendicular to the z direction, the force is proportional to the velocity gradient and is the coefficient of viscosity, η :

$$\sigma_{zx} = \eta \frac{\partial u_x}{\partial z}.$$

We will use kinetic theory to calculate the coefficient of viscosity.

The number of molecules in a fluid crossing a surface of unit area located in the plane $z = z_0$, and travelling in the $+z$ direction, is

$$\int_+ d\mathbf{v} v_z f(\mathbf{v}),$$

while the number of molecules crossing the surface in the opposite direction is

$$- \int_- d\mathbf{v} v_z f(\mathbf{v}).$$

- (a) Show that if the distribution of velocities in the gas is given by a Maxwell–Boltzmann distribution, then both of these expressions are equal to

$$\frac{n_0 \langle v \rangle}{4}.$$

- (b) Now consider the momentum carried between different layers of the fluid by the two sets of molecules (recalling that there is a velocity gradient in the z direction). Assume that the molecules crossing the surface in the $+z$ direction originate on average from $z = z_0 - l$, where l is the mean free path (the average distance between collisions of molecules). Hence show that the momentum transfer through the surface due to these molecules is

$$\frac{1}{4} \rho \langle v \rangle u_x|_{z=z_0-l}.$$

- (c) Give the expressions for the momentum transferred by molecules travelling in the $-z$ direction and the net momentum transfer through the surface in the $+z$ direction.
- (d) The net rate of momentum transfer found in parts (b) and (c) is equivalent to a force per unit area exerted in the x direction by the gas below $z = z_0$ upon the gas above this surface, which is given by σ_{zx} . Assuming that l is small on the scale of the experiment, find an expression for η .

5.7 You are on one side of a room in which the air is still, about 5 m from the far wall when hydrogen sulphide, H_2S (“rotten egg gas”) starts to leak in through a hole in the far wall. Estimate the time until you start to smell the poisonous gas.

5.8 Spectral lines from a hot body of gas may be broadened through the Doppler effect. For a spectral line which has wavelength λ_0 for molecules at rest, show that the combination of the motion of molecules in the gas and the Doppler effect leads to an intensity profile at wavelengths λ near λ_0 of

$$I(\lambda) \propto \exp \left[-\frac{mc^2(\lambda - \lambda_0)^2}{2\lambda_0^2 k_B T} \right].$$

This is known as thermal Doppler broadening of a spectral line.